

SOC ANALYST LEVEL 1

CYBER KILL CHAIN & MITRE ATT&CK

Mastering the sequence of intrusions and the unified language of adversary behavior.

Author: **Nader Anam** // Duration: **3 Hours**

| LEARNING OBJECTIVES

- Understand the **Cyber Kill Chain** stages.
- Differentiate Tactic, Technique, and Procedure.
- Map security behaviors to the **MITRE matrix**.
- Transform isolated alerts into a **Timeline**.
- Identify stages: Access, Execution, Persistence.
- Link Phishing, PowerShell, and C2 in a chain.
- Write professional verdicts using MITRE language.
- Make logical escalation decisions based on stages.

WHY THIS LECTURE MATTERS

A weak analyst sees an alert as a singular data point. A pro sees it as a **step in a story**.

"The incident is not a snapshot; it's a sequence."

By understanding the Kill Chain, you predict the attacker's next move and find evidence they tried to hide.



THE ISOLATED ALERT TRAP

THE ROOKIE VIEW

"I see a **Suspicious PowerShell** alert. PowerShell is encoded. Case closed."

THE PRO VIEW

"Encoded PowerShell. Why? Was there Phishing?
Did it call a C2? Are there internal login attempts?
I must connect the dots."

PART 01:

THE CYBER KILL CHAIN

The Linear Path of a Cyber Attack

CONCEPT OF THE KILL CHAIN

Developed by Lockheed Martin, this model breaks down an intrusion into **7 sequential stages**. If the defender breaks even ONE link in this chain, the attack fails. This is the foundation of **Defensive Architecture**.

in: The 7 Stages of a Cyber



THE 7 STAGES (1 - 4)



RECON

Gathering info on the target:
emails, IPs, software stack.



WEAPONIZE

Coupling exploit with backdoor
into a deliverable (e.g., infected
PDF).



DELIVERY

Transmitting the weapon to the
victim via email, link, or USB.



EXPLOIT

The malicious code executes on
the target system.

THE 7 STAGES (5 - 7)



INSTALL

Establishing a persistent foothold (backdoor, Registry keys).



C2

Command & Control: Host calls home to the attacker for orders.



ACTIONS

Achieving the mission: Data exfil, encryption, or sabotage.

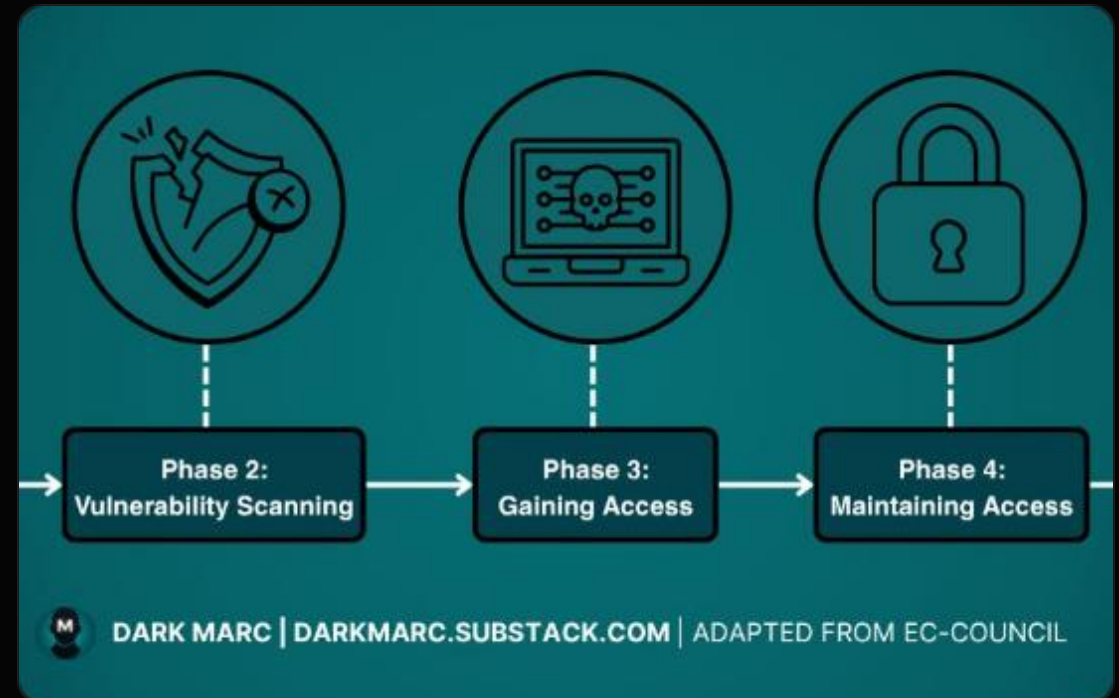
STAGE 1: RECONNAISSANCE

Finding the path of least resistance. Attackers use

OSINT and active scanning to map the target.

- Employee names on LinkedIn.
- Public IP range scanning.
- Software version fingerprinting.

SOC Perspective: Hard to see directly, but explains targeted phishing themes.



STAGE 2: WEAPONIZATION

PREPARING THE PAYLOAD

Creation of a malicious artifact. The attacker combines their malicious intent with a delivery vehicle.

- Malicious Office documents (Macros).
- Compiled EXEs (Trojans).
- Infected software updates.

EXAMPLES:

```
invoice.docm
```

```
VPN_Client_Update.exe
```

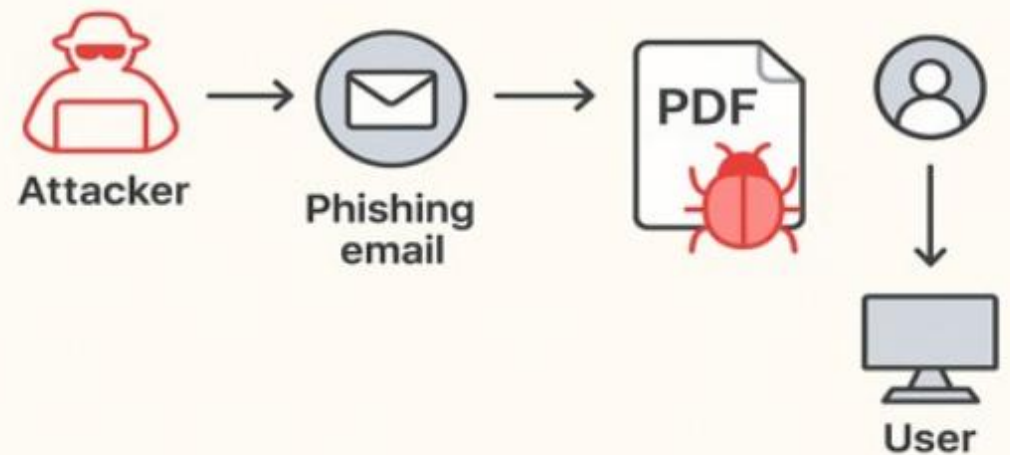
```
secure_login.html
```

STAGE 3: DELIVERY

The "Transmission" phase. This is the first link where SOC has high visibility via gateways.

- **Phishing:** Attachment or URL.
- **Web:** Drive-by downloads.
- **Physical:** Malicious USB drives.

SOC Marker: Email Gateway logs, Proxy history.



STAGE 4: EXPLOITATION

The payload is triggered. The attacker leverages a vulnerability in software or **human behavior**.

- User opens doc and clicks "Enable Content".
- Exploiting a zero-day in a browser.
- Brute-forcing an exposed service.

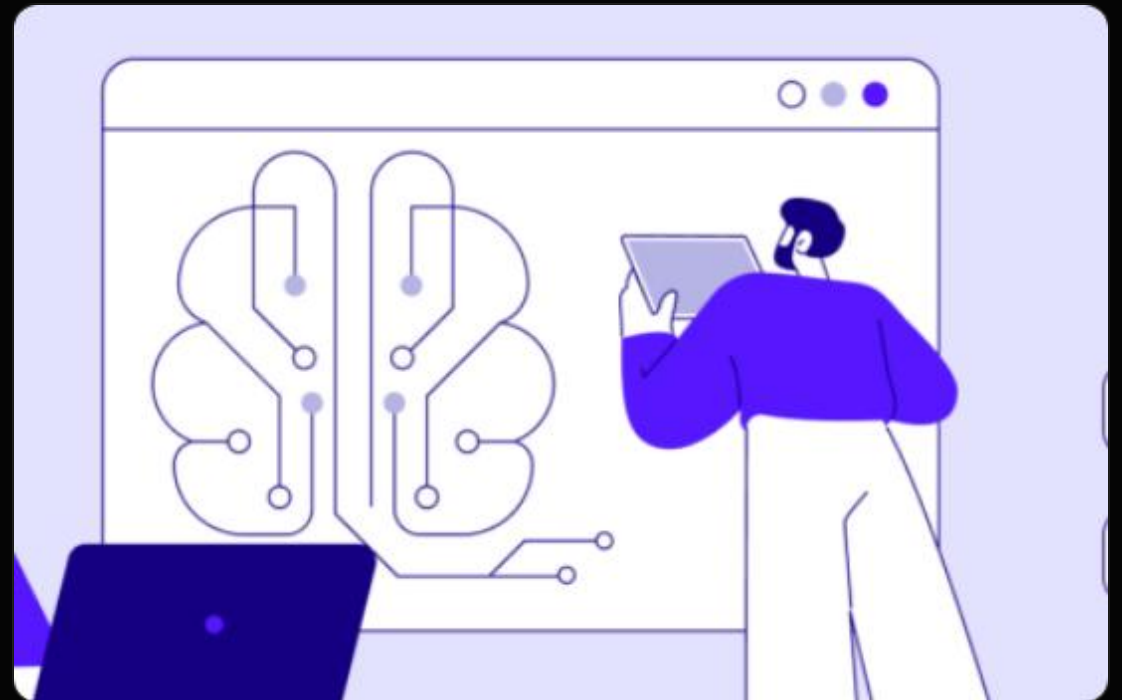
```
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VYiASPgQjN1NXyiRiCNoQDz5GZoNXdsZ2LgcWam5
```

STAGE 5: INSTALLATION

Establishing the **Foothold**. The goal is to remain in the system even if it reboots.

- Creating a persistent **Scheduled Task**.
- Modifying **Registry Run Keys**.
- Injecting code into system processes.

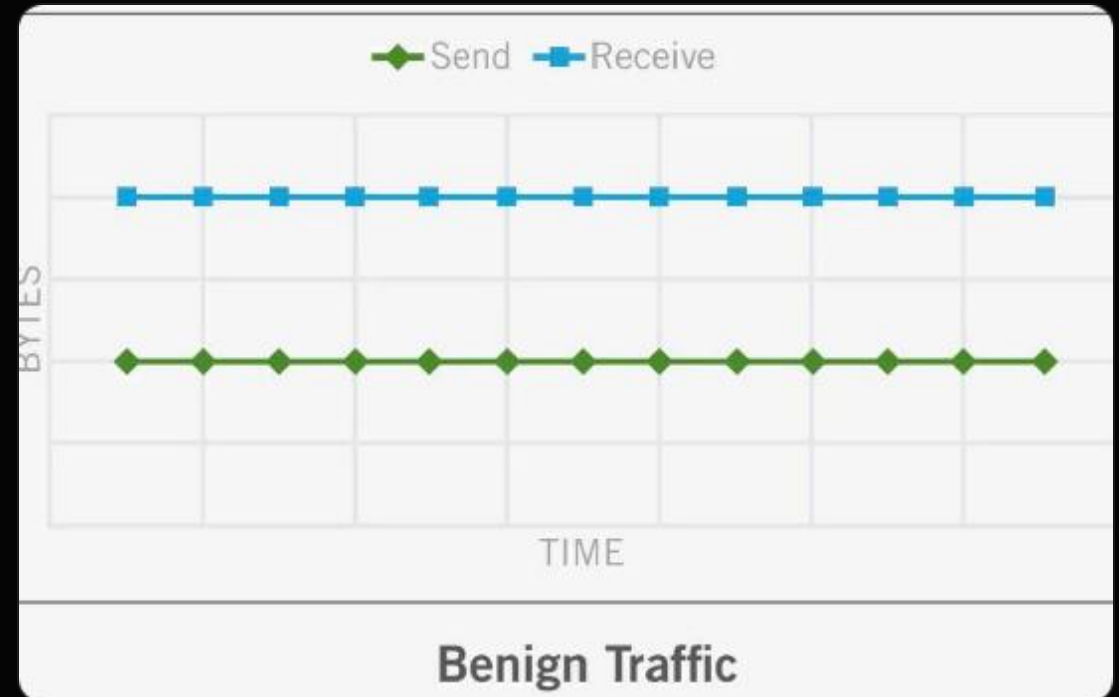
PERSISTENCE CHECK REQUIRED



STAGE 6: COMMAND & CONTROL

The attacker now has remote control. The host beacons to an external IP to receive orders.

- Periodic heartbeat (e.g., every 60s).
- Unusual outbound traffic on common ports (443/80).
- Domain queries to suspicious DNS nodes.



STAGE 7: ACTIONS ON OBJECTIVES

The mission phase. The attacker does what they came for:

- **Data Exfiltration:** Stealing files.
- **Ransomware:** Encrypting data.
- **Sabotage:** Wiping servers.

Severity: CRITICAL - Immediate Response!

YOUR COMPUTER
HAS BEEN LOCKED!

All your files have been encrypted.

This computer lock is aimed to stop your illegal activity.

To unlock your computer you are obliged to pay a fine.

You have 72 hours to pay the fine, otherwise you will be arrested.



OK

| LIMITATIONS OF THE KILL CHAIN

NON-LINEARITY

Attackers skip stages or loop back frequently.

GRANULARITY GAP

It doesn't explain HOW a task is created or HOW data is stolen.

INSIDE-OUT ATTACKS

Poorly describes insider threats or stolen credential usage.

To solve this, we use the **MITRE ATT&CK Framework**.

PART 02:

MITRE ATT&CK

The Global Knowledge Base of Adversary Behavior

WHAT IS MITRE ATT&CK?

MITRE ATT&CK is not a tool; it's a globally accessible knowledge base of adversary **Tactics** and **Techniques** based on real-world observations.

Standardizes the "Who, What, How" of an attack.

Resource Development 8 techniques	Initial Access 10 techniques	Execution 14 techniques	Persistence 20 techniques	Privilege Escalation 14 techniques	Defense Evasion 43 techniques	Credential Access 17 techniques
Acquire Access	Content Injection	Cloud Administration Command	Account Manipulation (6)	Abuse Elevation Control Mechanism (6)	Abuse Elevation Control Mechanism (6)	Adversary-in-the-Middle (3)
Acquire Infrastructure (8)	Drive-by Compromise	Command and Scripting Interpreter (10)	BITS Jobs	Access Token Manipulation (3)	Access Token Manipulation (3)	Brute Force (4)
Compromise Accounts (3)	Exploit Public-Facing Application	Container Administration Command	Boot or Logon Autostart Execution (14)	Account Manipulation (6)	BITS Jobs	Credentials from Password Stores (6)
Compromise Infrastructure (8)	External Remote Services	Deploy Container	Boot or Logon Initialization Scripts (5)	Boot or Logon Autostart Execution (14)	Build Image on Host	Exploitation for Credential Access
Develop Capabilities (4)	Hardware Additions	Exploitation for Client Execution	Browser Extensions	Boot or Logon Initialization Scripts (5)	Debugger Evasion	Forced Authentication
Establish Accounts (3)	Phishing (4)	Inter-Process Communication (5)	Compromise Host Software Binary	Create or Modify System	Deobfuscate/Decode Files or Information	Forge Web Credentials (2)
Obtain Capabilities (7)	Replication Through	Native API	Create	Domain or Tenant Policy Modification	Deploy Container	Input

CKC VS. MITRE ATT&CK

Cyber Kill Chain	MITRE ATT&CK
High-level stages (Recon, Delivery)	Granular behavior (PowerShell, Phishing)
Rigid and Linear sequence	Flexible matrix of objectives
Best for Strategic defense	Best for Tactical SOC Analysis & Hunting
Linear Flow focus	Attacker Behavior focus

UNDERSTANDING TTPS

TACTICS

The "Why" - The technical goal (e.g., Persistence).

TECHNIQUES

The "How" - The method to achieve the goal (e.g., Scheduled Task).

PROCEDURES

The "What" - The specific implementation (e.g., cmd.exe /c update.exe).

EXAMPLE: TTP IN ACTION

```
User opens invoice.docm → winword.exe spawns powershell.exe -enc ...
```

MAPPING:

Tactic: Execution

Technique: Command & Scripting Interpreter /
PowerShell

PROCEDURE:

Phishing-delivered Office doc launching Base64
PowerShell payload.

ESSENTIAL TACTICS FOR L1

- ➡ Initial Access
- > Execution
- 🕒 Persistence
- 📊 Privilege Escalation

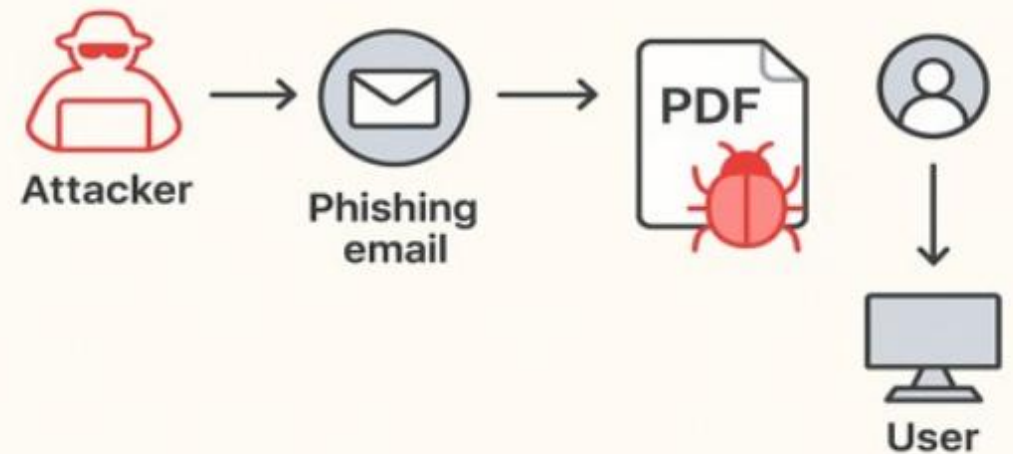
- 👻 Defense Evasion
- 🔑 Credential Access
- 🔍 Discovery
- 🚶 Lateral Movement

TACTIC: INITIAL ACCESS

The entry point. How the attacker gets in.

- Phishing (Link/Attachment).
- Exploiting external services (VPN/Web).
- Using Valid Accounts (Stolen creds).

SOC: User clicked a link & logged in.



TACTIC: EXECUTION

Running code. This is where "static" threats become "active."

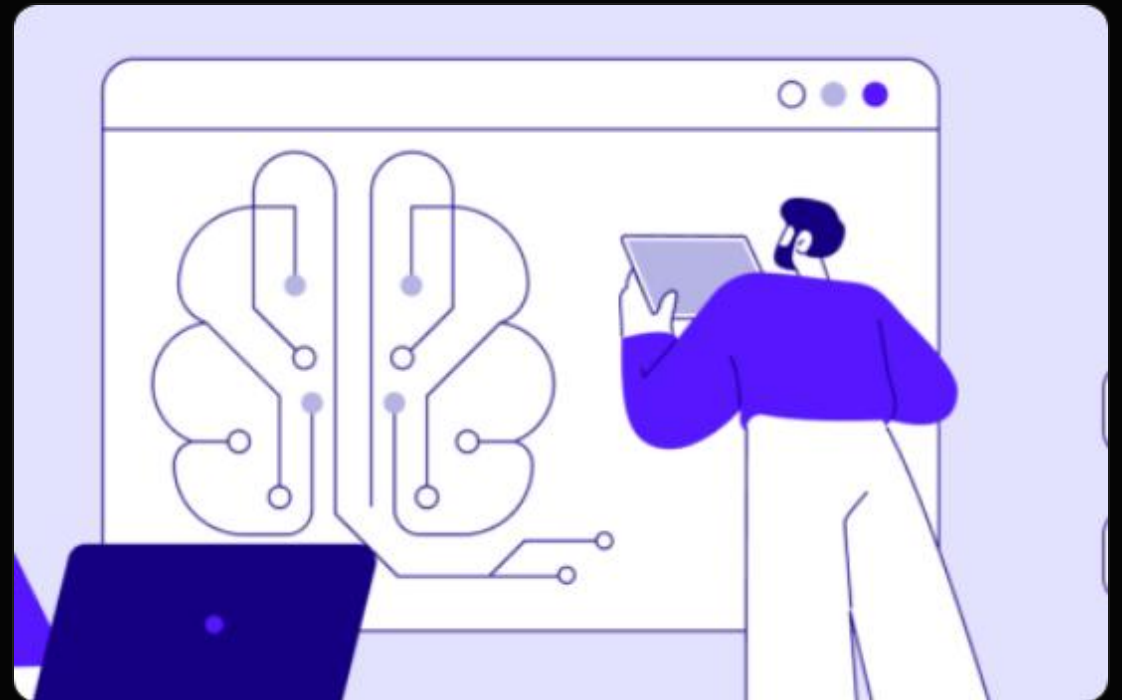
- PowerShell scripts (.)`ps1`
- Command line binaries (.)`cmd.exe`
- Macro execution in Office.

```
none;font-size: 14px;cursor: pointer;border-color 0.3s ease;}.second-button:hover>  
gcG93ZXJzaGVsbCAkSUEgPSAnYzpcdXN1cnNccHV  
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VYiASPgQjN1NXyiRiCNoQDz5GZoNXdsZ2LgcWam5
```


TACTIC: PERSISTENCE

The "Survival" tactic. Ensures access through restarts.

- New Scheduled Task.
- New Service entry.
- New Admin User account.
- Linux Cron Job.



TACTIC: PRIVILEGE ESCALATION

GOAL: ROOT/ADMIN

Moving from a standard user to a high-privilege account.

Event ID: 4672 (Special Privs)

INDICATORS:

- User added to 'Administrators' group.
- Usage of `sudo su`.
- Exploiting local kernel vulnerabilities.

TACTIC: DEFENSE EVASION

Hiding activity from SOC Analysts. **Critical Red Flag.**

CLEARING LOGS

Event ID 1102 / history -c

OBFUSCATION

Encoded PowerShell /
Renaming files

DISABLING SECURITY

Turning off Antivirus or EDR
sensors

TACTIC: CREDENTIAL ACCESS

Stealing passwords and tokens.

- Credential Dumping (LSASS).
- Phishing for credentials.
- Reading browser passwords.

SOC ALERT:

Process `mimikatz.exe` detected accessing system memory.

TACTIC: DISCOVERY

Attacker researching the infected system to see where they are.

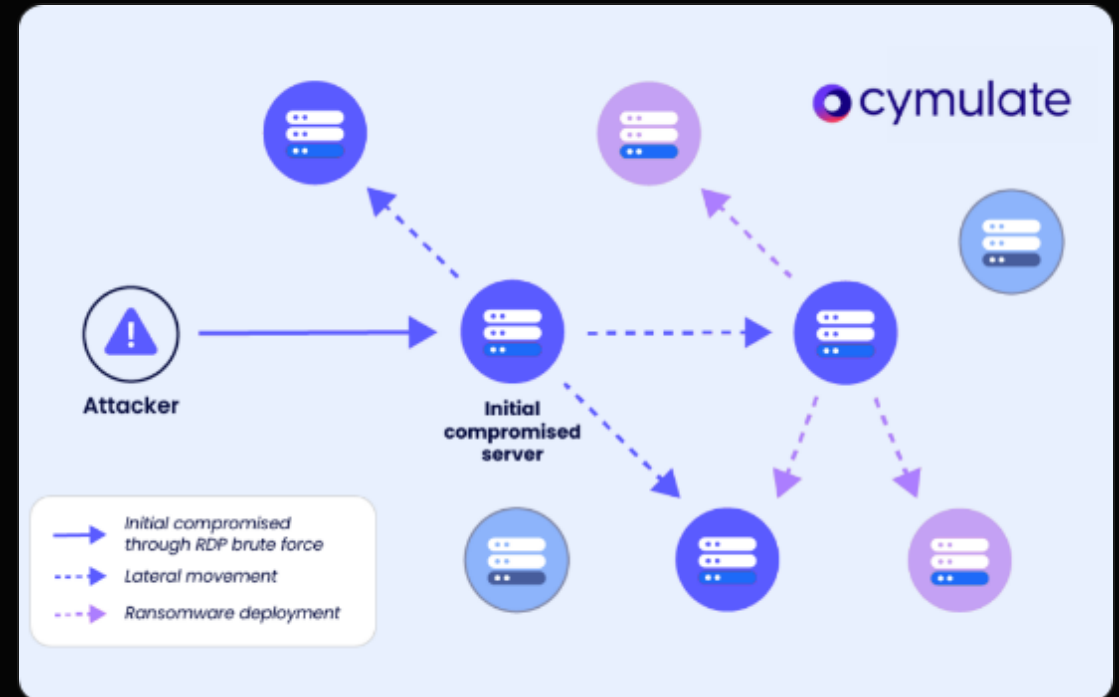
```
whoami | hostname | ipconfig | net user | net localgroup administrators
```

Repetitive recon commands after a suspicious login = **Discovery**.

TACTIC: LATERAL MOVEMENT

Moving from host to host. Attacking the **Internal Network**.

- Remote Desktop (Logon Type 10).
- SMB/Admin Shares.
- SSH to another Linux server.



TACTIC: COMMAND & CONTROL

ESTABLISHING REACH

Connections every 30/60 seconds (Beaconing).

Tiny, repetitive data bursts to an unknown IP.

SUSPICION:

10.0.0.25 → 91.200.x.x:443 (Freq: 60s)

EXFILTRATION & IMPACT

EXFILTRATION

Moving organizational data to an external cloud or server.

`Archive.zip uploaded to Mega.io`

IMPACT

Encryption, destruction, or denial of service.

`Ransomware message on desktop.`

DRILL 1: CLASSIFICATION

Match the security event to the MITRE Tactic:

Security Event	MITRE Tactic
Clearing Event Logs	?
PowerShell encoded command	?
New Scheduled Task	?
Login from A to B via RDP	?
Ransom Note found	?

DRILL 1: SOLUTION KEY

Security Event	MITRE Tactic
Clearing Event Logs	DEFENSE EVASION
PowerShell encoded command	EXECUTION
New Scheduled Task	PERSISTENCE
Login from A to B via RDP	LATERAL MOVEMENT
Ransom Note found	IMPACT

DRILL 2: ORDERING THE CHAIN

Order these events chronologically to build the story:

1. Internal RDP probes to another host.
2. User opens docm attachment.
3. Encoded PowerShell connects to external IP.
4. New Scheduled Task created.
5. Phishing email delivered.

DRILL 2: SOLUTION KEY

1. Phishing delivered (Initial Access)
2. User opens docm (Execution)
3. PowerShell C2 (Command & Control)
4. Scheduled Task (Persistence)
5. RDP probes (Lateral Movement)

FULL SCENARIO: CASE MONA

```
Subject: Urgent Invoice Review  
Attachment: invoice.docm  
winword.exe → powershell.exe -enc ...  
10.0.0.25 → 45.77.x.x every 60s  
New Task created: "update_check"  
Failed logins to FILE-SRV-01
```

CASE MONA: TIMELINE

T0

Phishing email delivered



T+2M

Macro Exec / PowerShell run



T+5M

C2 Beacons established



T+15M

Foothold via Task created



T+30M

Lateral Move probes to server



CASE MONA: VERDICT

SEVERITY: CRITICAL

Verdict: Verified compromise. Initial access via phishing led to malware execution and active C2 beaconing. Attacker is now attempting lateral movement.

Action: ESCALATE IMMEDIATELY. Revoke account, Isolate host.

PROFESSIONAL REPORT TEMPLATE

The activity indicates a phishing-led intrusion.

- Tactic: Initial Access (Phishing)
- Tactic: Execution (PowerShell)
- Tactic: C2 (Encrypted Beacons)
- Tactic: Persistence (Scheduled Task)

Escalation required.

| COMMON MITRE MISTAKES

- Mapping without evidence.
- Confusing Technique with Tactic.
- Ignoring the timeline.
- Forcing an alert to fit a tactic incorrectly.

| THE GOLDEN RULE



CONNECT THE STORY TO THE LOGS.

Never write MITRE mapping without a Timestamp and Evidence Log ID.

POP QUIZ

Q1: Phishing is which MITRE Tactic?

A) Initial Access | B) Discovery

Q2: Deleting logs is which Tactic?

A) Defense Evasion | B) Impact

Q3: Moving between hosts is?

A) Lateral Movement | B) Collection

QUESTIONS?

Thank you for your attention.

Next Session: Alert Triage Methodology

IMAGE SOURCES



https://miro.medium.com/v2/resize:fit:1400/1*Jav5T72Ad7xHhGFsJjip0w.png

Source: medium.com



https://miro.medium.com/1*ZOJcL8t27dirxRFECNong.png

Source: ibrahimakkdag.medium.com



[https://substackcdn.com/image/fetch/\\$s_!-5xJ!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2Fae954006-3afe-48ec-807c-0c51341e0d91_1500x500.png](https://substackcdn.com/image/fetch/$s_!-5xJ!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2Fae954006-3afe-48ec-807c-0c51341e0d91_1500x500.png)

Source: darkmarc.substack.com



<https://blog.cleanfox.io/wp-content/uploads/2025/06/malware-attachment-phishing-email.jpg>

Source: blog.cleanfox.io



https://image.ahnlab.com/atip/content/atcp/2024/05/4_3.png?w=704&ssl=1

Source: asec.ahnlab.com



<https://cymulate.com/uploaded-files/2024/05/img-8-1024x314.png>

Source: cymulate.com

IMAGE SOURCES



https://cdn.prod.website-files.com/64e50cbe2b6f932c04238c14/651bd76044e4edfe8ad0799b_6502f38429ffb1cdade0bc0b3_example-benign-traffic.webp

Source: www.vectra.ai



<https://www.dataprise.com/wp-content/uploads/2022/11/YOUR-COMPUTER-HAS-BEEN-LOCKED-1024x576.png>

Source: www.dataprise.com



https://miro.medium.com/v2/resize:fit:1400/1*kLBMiDCX7Ac0LTk2CnXrWw.jpeg

Source: medium.com



<https://cymulate.com/uploaded-files/2024/11/lateral-movement-1024x593.png>

Source: cymulate.com